

A Researcher wishes to estimate the average of an attribute by using sample technique with 93% confidence and error not more than 2. If population SD is 22, Compute the appropriate sample size.

Solution:-We have,

$$\begin{aligned}\text{Confidence level}(1-\alpha) &= 0.93 \\ \alpha &= 0.07 \\ Z\alpha &= 1.811911 = \text{NORMSINV}(1-0.07/2) \\ \text{Max. permissible error (E)} &= 2 \\ \text{Pop. S.D.}(\sigma) &= 22\end{aligned}$$

$$\text{Now, Required Sample (n)} = 397.2455 = (E\sigma/Z\alpha)^2$$

Hence, Required sample size is 398.

Name:-Hari Bashyal